

# *Biology*

## **Chapter 1: Cell Structure and Function**

# Welcome to AnithUncommon's Sample Resources!

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We are assuming you are reading this as a student. So What will **YOU** Expect From This Sample?

*In Biology, Cell Structure and Function, Ch. 1.1*

This sample will be divided into **three** parts: Introduction to Samples (this part), Lesson Objectives (What is Expected From Students), and most crucially, Syllabus Contents.

Note that the 'Syllabus Content' does NOT stand as a complete syllabus itself; rather, it is a tight compilation of notes gathered by Azalia to create a comprehensive sample for students.

If you want a complete version of these notes and think that you need help in Biology, join our nonprofit!

*Gmail: [anithuncommon](#)*

*Instagram handle: [@anithuncommon](#)*

This sample is made thanks to Azalia.

*— Made by students, for students.*

## Lesson Objectives:

At the end of the lesson, students should be *able* to...

1. **Recognize** and **understand** the function of organelles and other cell structures found in eukaryotic cells
2. **Compare** the structure of typical plant and animal cells
3. **Outline** and **understand** key structural features of a prokaryotic cell as found in a typical bacterium
4. **Compare** the structure of a prokaryotic cell as found in a typical bacterium with the structures of typical eukaryotic cells in plants and animals

## Reflective Question to Consider:

A cell is the basic unit of life, and all organisms are composed of one or more cells. The two fundamental types of cells are prokaryotic and eukaryotic. They have their own roles and functions, beside the fact that they share some organelles.

Below is the following core questions you will face as we explore this chapter:

- Does the choice between rigid cell wall and a flexible membrane actually reflect the way cells cope with their environment?
- Why do eukaryotes rely on a specialized system of organelles, while prokaryotes do it all in an open space?

# Unit 1: Cell Structure and Function

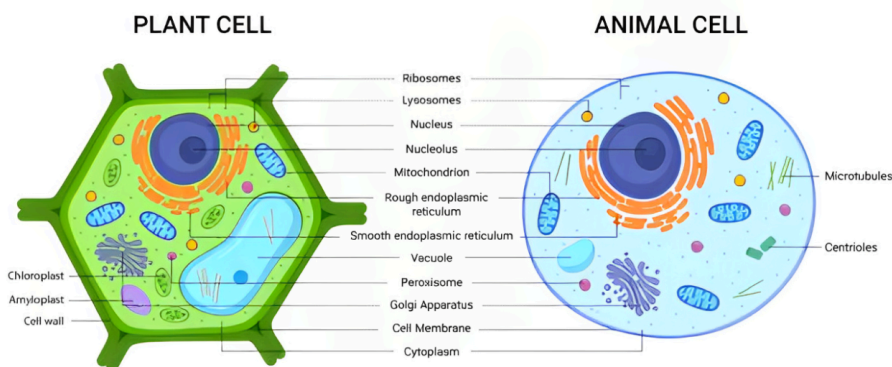
## Pro Tip!

- Memorizing cell organelles might be a tricky task, but you can draw them to memorize them easier. Understanding how they look also helps in memorizing the functions.

## Activities in Class:

- Lesson Breakdown
- Gimkit-Based Review for deep memorizing
- Collaborative cell drawing using Draw Chat

## 1.1 Two Types of Cells



## Eukaryotic Cell

### A. Structure and Function

#### a. Cell surface membrane

→ To protect the **outermost part** of the cell, prevent **cytoplasm** from **spreading out** and control the **in-and-out movement** of the cell (more like a thin layer filtering the size of molecules passing through).

#### b. Nucleus, nuclear envelope and nucleolus

→ Nucleus is the 'brain' that **controls all activities** throughout the cell besides acting as a place where

## Mentor's Notes ☐★

It's actually pretty complicated to remember all of the cell organelles at once, so I provide lyrics to remember the whole animal cell organelles. I got this from my prev bio teacher as she sang this with "Twinkle, Twinkle Little Stars" melody:

*Nucleus is control  
brain,  
DNA in its domain.*

**DNA is stored.**

→ Nucleolus is a small region in the nucleus that is responsible for **ribosome making**.

→ Nuclear envelopes **protect** both nucleus and nucleolus, also all that are contained inside.

**c. Rough endoplasmic reticulum**

→ 'Rough' here resembles the **uneven, bumpy appearance** of RER since it's filled with ribosomes.

→ RER primarily acts as the **protein chain producer** and modifiicator. It also **transports** them to golgi apparatus.

**d. Smooth endoplasmic reticulum**

→ While RER is bumpy, SER has a more **even appearance** due to the lack of ribosomes present.

→ SER is involved in **lipid metabolism** and storage.

**e. Golgi apparatus**

→ Is a stack of **flattened cisternae** (membrane-bound sacs) that's kinda similar to SER but with different functions.

→ Golgi apparatus functions as a **lysosome**—vesicles containing digestive enzymes—producer and **lipid transporter**.

**f. Mitochondria (s: mitochondrion)**

→ "The powerhouse of cells"

→ The primary site of **ATP production** through oxidative phosphorylation.

→ Has its own mitochondrial **DNA** and **70S ribosomes**.

**g. Ribosomes**

→ Mini, freeflying site of **protein-making**, especially **transcription and translation**.

→ Composed of **2 subunits**: large subunit and small subunit.

*Ribosomes make protein chains, ER moves them like some trains.*

*Golgi packs and ships with care, Lysosomes clean everywhere.*

*ATP in mitoci, Ribosomes in their 80-s.*

*Vacuoles store stuffs inside, Cell membrane is outer guide.*

*Let's all learn about the cell, 1,2,3, it rings a bell!*

*I hope it's helping to remember, but you still have to understand the details bcs it's way too simple hehe.*

*Drawing can actually help a lot in understanding the*

→ Some move **freely in cytoplasm** while some are **bound to RER**.

#### **h. Lysosomes**

→ **Single-membraned sacs** containing digestive enzymes, released from golgi apparatus.

→ Useful in **breaking down waste** and old organelles.

#### **i. Microtubules, centrioles, and centrosomes**

→ Microtubules are slim tubes that **provide structure** to the cell, keeping it hollow inside.

→ Centriole is a star-shaped organelle made of **nine triplets of microtubules**.

→ Centrosome is a **pair of centrioles**. Has a prominent role during **spindle fiber formation** in cell division.

→ All of them are present in **animal cells only**.

#### **j. Cilia**

→ A **hair-like structure** outside the cell, used to **control movements** and cope with specific situations.

→ An example can be found on the surface of specific epithelial cells, such as those lining the **trachea, bronchi, and bronchioles**.

#### **k. Microvili**

→ An **elongation** of the cells, used to **expand the surface area**.

→ Specifically found in the **small intestine** to maximize absorption.

#### **l. Chloroplast**

→ Another powerhouse but specifically found **only in plant cells**.

→ Contains **chlorophyll**, the pigment that colors plants **green**.

→ Is a primary site for **ATP and starch production** as well as storage.

cell's organelles. If you are taking notes by hand about this subtopic, it's better to make drawings of the structure to help deepen your learning.

There's a unique theory in Campbell's Book that may shift your perspective in seeing the world of prokaryotes and eukaryotes: The Endosymbiotic Theory. The theory states that eukaryotic organelles, specifically mitochondria and chloroplasts, originated as free-living prokaryotic microbes that are "eaten" by an euka host cell. That's why prokaryotes have similar characteristics to

**m. Cell wall**

→ A rigid wall composed of either **cellulose or chitin (fungi)** used to line the outer part of the **plant cell only**.

**n. Plasmodesmata**

→ A **junction** that connects between cells, lined through the cell walls.

→ Found only in **plant cells**.

**o. Large permanent vacuole and tonoplast**

→ Vacuole is a **big 'lake'** in the plant cells containing **water supply and glucose**. This is also present in animal cells but in a smaller version.

→ Tonoplast is a **thin membrane** engulfing the vacuole.

**B. Animal Vs Plant Cell**

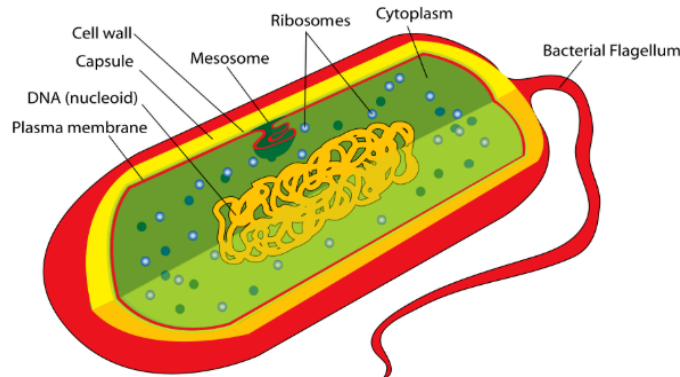
\*Note: If the organelles/features aren't listed below, it means they are present in both animal and plant cells.

| The Difference of Plant Cell and Animal Cell |                        |                          |
|----------------------------------------------|------------------------|--------------------------|
| Feature                                      | Plant Cell             | Animal Cell              |
| Cell wall                                    | ✓                      |                          |
| Chloroplast                                  | ✓                      |                          |
| Plasmodesmata                                | ✓                      |                          |
| Centriole                                    |                        | ✓                        |
| Lysosome                                     |                        | ✓                        |
| Microvili & Cilia                            |                        | ✓                        |
| Vacuole                                      | Large, central vacuole | Small, temporary vacuole |

mitoci and chloroplasts, such as 70S ribosomes, almost the same size and producing their own ATP. You can remember it that way to compare between proka and euka.

|       |              |                  |
|-------|--------------|------------------|
| Shape | Rigid, fixed | Irregular, fluid |
|-------|--------------|------------------|

## Prokaryotic Cell



### A. Key Features and Structure

#### a. Unicellular

→ **One cell** as a **whole organism**. Unlike eukaryotes that need many cells to build up systems, this one only needs a single cell as the whole thing.

#### b. Generally 1-5 $\mu\text{m}$ diameter

→ This size is equal to mitochondrion, making it roughly **10-100 times smaller** than eukaryotics.

#### c. Cell walls

→ Made of **peptidoglycan**, used to prevent prokaryotes from **bursting** under certain conditions.

#### d. Circular DNA

→ Called '**plasmid**', useful in microbiology studies and **genetic expression**.

→ Beside plasmid, there is also the **central DNA**. The central DNA is generally loose, **floating inside** the cytoplasm—unlike eukaryotes that have nuclei.

#### e. 70S ribosomes

→ The same ribosome size as **mitochondria** in eukaryotics.

#### f. Absence of organelles surrounded by double membranes



→While most organelles in eukaryotes (nuclei, chloroplast, mitochondria) are **bilayered**, this one **doesn't**. They lack these structures to keep the **simplicity** and **smaller in size**.

## B. Prokaryotic Vs Eukaryotic

| The Difference of Prokaryotic Cell and Eukaryotic Cell |                               |                                    |
|--------------------------------------------------------|-------------------------------|------------------------------------|
| Feature                                                | Prokaryotic                   | Eukaryotic                         |
| Size                                                   | 1-5 $\mu\text{m}$             | 10-100 $\mu\text{m}$               |
| DNA                                                    | Circular and irregular        | Linear, tied by histone proteins   |
| Ribosome                                               | 70S                           | 80S                                |
| Cell wall                                              | Peptidoglycan                 | Cellulose (plants), chitin (fungi) |
| Organelles                                             | No double-membraned organelle | Membrane-bounded organelle         |
| Division                                               | Binary fission                | Mitosis & meiosis                  |
| Cellular system                                        | Unicellular                   | Multicellular                      |

# Mentors' Sources:

Video link to make a clearer visualization of cells:

[Cell Biology](#)

Campbell 12th Edition:

[Campbell 12th Edition](#)

SaveMyExams Revision Notes:

[SaveMyExams](#)